

DATA DRIVEN METHODS TO PREDICT TURBULENT/EDDY VISCOSITY

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ABSTRACT

Turbulence modeling remains one of the major challenges in computational fluid dynamics (CFD) due to the high computational cost and limited generalizability of conventional turbulence closure models. In the present work, a data-driven framework based on the Sparse Identification of Nonlinear Dynamics (SINDy) algorithm is implemented for the prediction of turbulent eddy viscosity and turbulence closure modeling in turbulent flows. The proposed methodology combines CFD-generated flow field data and high-fidelity DNS datasets with sparse regression techniques to identify compact and interpretable governing relations between important turbulence quantities. Flow variables such as turbulent kinetic energy k , turbulent dissipation rate ϵ , velocity gradient components ∇U , and Reynolds stress quantities are utilized to construct nonlinear candidate libraries for the SINDy framework.

Initially, CFD data generated from OpenFOAM simulations for flow past a cylinder using the $k - \omega$ SST turbulence model are employed for prediction of turbulent viscosity and comparison against classical turbulence models. The SINDy algorithm identifies the most significant nonlinear terms contributing to turbulent viscosity prediction while eliminating redundant features through sparse optimization.

Additionally, a transferability study is performed using DNS datasets, where sparse turbulence closure relations are discovered from transitional boundary layer flow data and subsequently transferred to square duct flow without re-training. The transferred model is evaluated by comparing DNS and SINDy predictions through contour visualization and error analysis.

The developed framework aims to reduce computational complexity while retaining physical interpretability compared to conventional machine learning approaches. The study demonstrates the potential of sparse system identification techniques for developing efficient reduced-order turbulence models and highlights the applicability of SINDy in turbulence closure discovery and transfer learning for CFD applications.

1. INTRODUCTION

1.1. Background

Computational Fluid Dynamics (CFD) has become an essential tool for analyzing complex fluid flow phenomena in engineering applications such as aerospace, automotive systems, turbomachinery, and renewable energy systems. Accurate prediction of turbulent flow behavior remains one of the most challenging aspects of CFD due to the nonlinear and multiscale nature of turbulence. Conventional turbulence models such as the $k-\epsilon$ and $k-\omega$ models rely on empirical assumptions and closure relations to approximate turbulent stresses and eddy viscosity. Although these models are computationally efficient and widely used, their predictive capability may be limited for complex flow conditions.

1.2. Objectives

$$\frac{\partial \bar{u}_i}{\partial x_i} = 0 \quad (1)$$

$$\rho \left(\frac{\partial \bar{u}_i}{\partial t} + \bar{u}_j \frac{\partial \bar{u}_i}{\partial x_j} \right) = -\frac{\partial \bar{p}}{\partial x_i} + \mu \frac{\partial^2 \bar{u}_i}{\partial x_j^2} - \rho \frac{\partial \overline{u'_i u'_j}}{\partial x_j} \quad (2)$$

Our objective is to solve the RANS Equation, more specifically by getting an expression for Reynold Stress Term. The Reynold Stress term has been a major headache for the fluid mechanics committee until Boussinesq present an approximation for the Reynold Stress term as this term was inducing turbulence closure problem. He suggested to approximate this term like Newton closure model, Stress is directly proportional to strain. The constant of proportionality will give rise to the new term in fluid mechanics Turbulent/ Eddy Viscosity.

In recent years, data-driven modeling techniques and machine learning approaches have emerged as promising alternatives for improving turbulence modeling and reducing computational cost. These methods utilize high-fidelity simulation or experimental data to identify hidden patterns and relationships between flow variables. However, many machine learning models operate

as black-box systems and often lack physical interpretability, making it difficult to understand the governing mechanisms behind the predictions.

The Sparse Identification of Nonlinear Dynamics (SINDy) framework provides an interpretable and efficient approach for discovering governing equations directly from data. Instead of relying on highly complex neural network architectures, SINDy identifies the most significant nonlinear terms contributing to the system dynamics through sparse regression techniques. This enables the construction of compact mathematical models that retain physical meaning while reducing unnecessary complexity.

In the present work, the SINDy algorithm is implemented for the prediction of turbulent eddy viscosity using CFD-generated datasets obtained from OpenFOAM simulations. Important turbulence quantities such as turbulent kinetic energy k , turbulent dissipation rate ϵ , and velocity gradient components ∇U are extracted from the simulation data and used as input features for the sparse identification framework. The objective of the study is to develop a reduced-order and interpretable model capable of predicting turbulent viscosity with reasonable accuracy while minimizing computational complexity. The proposed methodology demonstrates the potential of sparse system identification techniques for future development of data-driven turbulence closure models in CFD applications.

1.3. Literature Review of SINDy (Sparse Identification of Non linear Dynamics) Algorithm

Discovering dynamical systems models from the data is a central challenge in mathematical physics, with a rich history going back at least as far as the time of Kepler and Newton and discovery of the laws of planetary motion. Historically, this process relied on a combination of high quality measurements and expert intuition. With vast quantities of the data and increasing computational power, the automated discovery of governing equation and dynamical systems is a new and exciting scientific paradigm.

Typically, the form of a candidate model is either constrained via prior knowledge of the governing equation, as in Galerkin projection or a handful of heuristic models are tested and parameters are optimized to fit data. Alternatively, best fit linear models may be obtained using DMD (Dynamic Mode Decomposition) or ERA (Eigenvalue Realization Algorithm). Simultaneously identifying the nonlinear structure and parameters of a model from the data is considerably more challenging, as there are combinatorially many possible model structure.

The SINDy algorithm (Brunton et al. 2016a) bypasses the intractable combinatorial search through all possible model structures, leveraging the fact that many dynamical systems,

$$\frac{d}{dt}x = f(x)$$

have dynamics f with only few active terms in the space of possible right hand side functions; for example, the Lorentz equation only have a few linear and quadratic interaction terms per equation.

We then seek to approximate f by a generalized linear model,

$$f(x) \approx \sum_{k=1}^p \theta_k(x) \zeta_k = \Theta(x) \zeta,$$

with the fewest possible non zero terms in ζ . It is then possible to solve for the relevant terms that are active in the dynamics using sparse regression, which penalizes the number of terms in the dynamics and scales well to large problems.

A library of candidate nonlinear functions $\Theta(x)$ may be constructed from the input data (in our case pressure, turbulent kinetic energy, production rate of turbulence, gradient of velocities, etc) which can also contribute to capturing the physics behind the phenomenon. Suppose the input data is represented by X then the library of non linear functions can be written as $\Theta(X) = [1, X, X^2, \dots, \sin(X)]$. The dynamical system may be represented in terms of the data matrices as

$$\dot{X} = \Theta(X)\eta$$

Each column ζ_k in η is a vector of coefficients determining the active terms in the k^{th} row in the above equation. A parsimonious model will provide an accurate model fit in the above equation with as few terms as possible in η . Such a model may be identified using a convex l-1 regularized sparse regression:

$$\zeta_k = \underset{\eta_k}{\operatorname{argmin}} ||\dot{X}_k - \Theta(X)\zeta_k||_2 + \lambda ||\zeta_k||_1$$

Here,

The result of the SINDy regression is a parsimonious model that includes only the most important terms required to explain the observed behavior. The sparse regression procedure used to identify the most parsimonious nonlinear model is a convex procedure. The alternative approach, which involves regression onto every possible sparse nonlinear structure, constitutes an intractable brute-force search through the combinatorially many candidate model forms. SINDy bypasses this combinatorial search with modern convex optimization and machine learning. It is interesting to note that for discrete-time dynamics, if $\Theta(X)$ consists only of linear terms, and if we remove the sparsity promoting term by setting $\lambda=0$, then this algorithm reduces to the DMD (Rowley et al. 2009, Schmid 2010, Tu et al. 2014, Kutz et al. 2016). If a least-squares regression is used, as in DMD, then even a small amount of measurement error or numerical roundoff will lead to every term in the library being active in the dynamics, which is nonphysical. A major benefit of the SINDy architecture is the ability to identify parsimonious models that contain only the required nonlinear terms, resulting in interpretable models that avoid overfitting.

The SINDy algorithm has recently been applied to discover model in fluid dynamics (El Sayed M et al. 2018, Loiseau & Brunton 2018). Because of the high state dimension, SINDy models of fluids are typically based on the POD (Proper Orthogonal decomposition) coefficients or low dimensional measurements such as lift and drag coefficients. Recent studies have also leveraged SINDy for turbulence modeling (Beetham & Capece de Nardis 2020, Schmelzer et al. 2020). SINDy has also been applied to identify models in non linear optics (Solo), chemical reactions dynamics, numerical algorithms, among others.

Because SINDy is formulated in terms of linear regression in a non linear library, it is highly extensible. The SINDy framework has been recently generalized by Loiseau and Brunton (Loiseau & Brunton) to incorporate known physical constraints

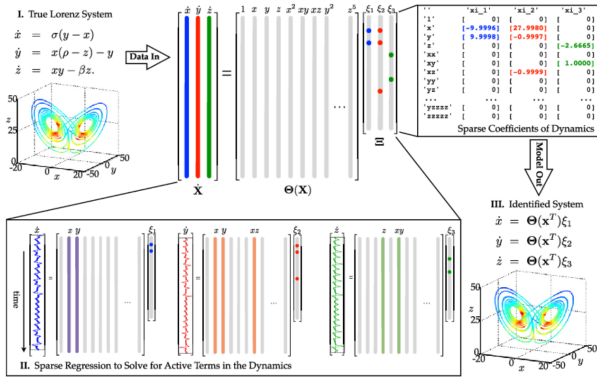


FIGURE 1: SINDy Architecture for Lorenz system

and symmetries in the equations by implementing a constrained sequentially thresholded least squares optimization. In particular, energy-preserving constraints on the quadratic nonlinearities in the Navier-Stokes Equation were imposed to identify fluid systems (Loiseau & Brunton 2018), where it is known that these constraints promote stability (Majda & Harlim 2012, Balajewicz et al. 2013, Carlberg et al. 2017). This work also showed that polynomial libraries are particularly useful for building models of fluid flows in terms of POD (Proper Orthogonal Decomposition) coefficients, yielding interpretable models that are related to classical Galerkin Projection (Brunton et al. 2016a, Loiseau & Brunton 2018). Loiseau et al. (2018) also demonstrated the ability of SINDy to identify dynamical system models for high-dimensional systems, such as fluid flows, from a few physical sensor measurements, such as lift and drag measurements on cylinder. For actuated systems, SINDy has been generalized to include inputs and control (Brunton et al. 2018), and these models are highly effective for model predictive control (Kaiser et al. 2018). It is also possible to extend SINDy algorithm to identify dynamics with rotational function nonlinearities (Mangan et al.), integral terms (Schaeffer & McCalla 2017), and based on highly corrupt and incomplete data (Tran & Ward 2016). SINDy was also recently extended to incorporate information criteria for objective model selection (Mangan et al. 2017), and to identify models with hidden variables using delay coordinates (Brunton et al. 2017). Finally, the SINDy framework was generalized to include partial derivatives, enabling the identification of partial differential equation models (Rudy et al. 2017, Schaeffer 2017).

2. METHODOLOGY

We are implementing SINDy algorithm for obtaining the turbulent viscosity which fluid mechanics committee rely on classical turbulence models like $k-\omega$ and $k-\epsilon$ model. So for our test case we will apply SINDy algorithm for flow past cylinder with a Reynold no of 63000 ($Re = 63000$) (highly turbulent) with inlet flow velocity of **1 m/s** and kinematic viscosity of 1.6×10^{-5} . The square cylinder has a side of 1 m. This highly turbulent flow has taken to capture the turbulent viscosity as lower turbulence will not significantly capture the turbulent viscosity. This will make sure to obtain an accurate dataset to train the SINDy algorithm. **Assuming steady state results for the analysis of flow past cylinder, hence the dataset obtained would be taken af-**

ter the steady state has been achieved. We have discretized the domain using **blockMesh** with multiple blocks (utility provided by OpenFOAM) and uniform meshing has been used all over the different blocks created. We have created total of 6 blocks of type **hex** to create the geometry. The simulations are performed by steady state solver **simpleFoam**. The results are then being analyzed by the **Paraview**, postprocessing software. For obtaining the $K-\omega$ SST data into csv (Comma Separated Values) files, we are using Paraview also analyzing SINDy (predicted) data in contours Paraview has been used.

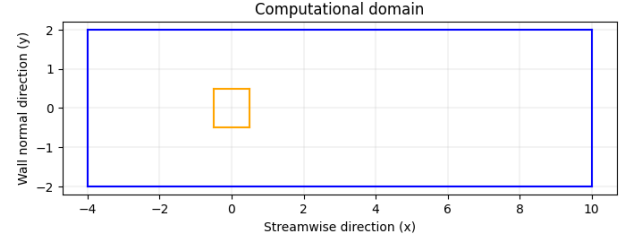


FIGURE 2: Computational Domain for flow past cylinder

The following image shows the contour plot for the velocity profile obtained by $K-\omega$ SST model. The alternating vortex shedding can be seen in the image which confirms that the flow is highly turbulent in nature. Mentioning about the boundary conditions, over the walls **no slip** boundary condition has been implemented and over the cylinder **slip** boundary condition has been applied.

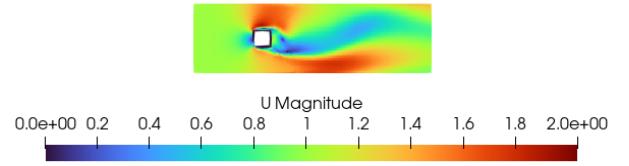


FIGURE 3: Velocity profile for flow past cylinder using $k-\omega$ SST model

After obtaining the dataset from existing turbulence model we will try to train SINDy algorithm by creating the non linear library of functions. Since SINDy is a linear regression of non linear library it must be a Supervised learning. Hence the target value for supervising the training of SINDy will be the turbulent Viscosity from the existed turbulence model ($K-\omega$ SST). Now we will implement SINDy using the **PySINDy** as SINDy has built in python library by Steven Brunton. The optimizer used to train this algorithm is 'STLSQ' which is included in the python library "PySINDy". The 'STLSQ' optimizer removes the weak terms from the non linear library which doesn't affect the physics significantly. Also a L1 regularization method will be effective to train the above algorithm

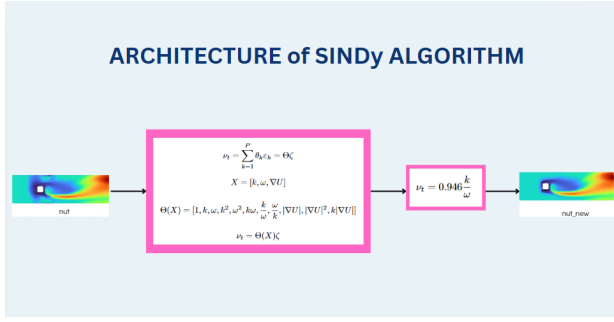


FIGURE 4: SINDy Algorithm Architecture

3. RESULTS AND DISCUSSION

3.1. Case Study I : Turbulent Eddy Viscosity Prediction using OpenFOAM and SINDy

The Turbulent/ Eddy Viscosity by the classical K- ω SST model is given by,

$$\nu_t = \frac{\rho k}{\max(a_1 \omega, SF_2)} \quad \text{where the } a_1 \text{ and } SF_2 \text{ represent a constant and Shear Force term respectively.}$$

The Turbulent/ Eddy Viscosity obtained by the SINDy algorithm is given by the expression,

$$\nu_t = 0.946 \frac{k}{\omega}$$

We are presenting results from SINDy in terms of turbulent viscosity along the different paths followed over the contours. As this will provide us an estimation about how the SINDy has achieved the prediction as compared to classical turbulence model (K- ω SST). First we will present results along the centerline of the cylinder both for SST and SINDy and we will compare them using scatter plot, see fig.5. Then we will analyze the results for an offset of 0.5 (cylinder upped side) and we will present how both models are varying along that offset of 0.5 over the contour, see Fig.6. At the end we will go and compare results from the both models along the diagonal of the computational domain and will present the variation through a scatter plot, see Fig.7. As a part of results we are also showing how the turbulent viscosity contours vary spatially since we have done steady state analysis, so temporal variation has been neglected as earlier mentioned in the methodology, see Fig.8. The mean square error for the K- ω SST and SINDy is coming out to be 4.6×10^{-7} . Also the R^2 score for the same is coming out to be 0.875.

We also have tried applying SINDy for a **DNS data for square duct** flow. The contour plot for DNS data and SINDy are given in Fig.9. The Mean Square Error for this square duct case is coming out to be 4.48×10^{-3} .

4. CONCLUSION

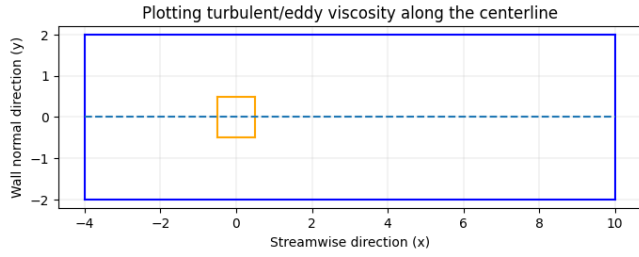
From the results obtained, it can be concluded that the predictions obtained from SINDy show good agreement with the classical $k - \omega$ SST turbulence model. In particular, along the diagonal of the computational domain, both models exhibit closely overlapping behaviour, indicating the capability of SINDy to capture the dominant turbulence characteristics. The obtained results demonstrate the effectiveness of sparse identification techniques for nonlinear dynamical systems and turbulence modeling.

An important observation is that SINDy was able to predict turbulent viscosity without explicit computation of terms such as the shear force contribution required by the SST model. Instead, the prediction relies upon the sparse governing relations extracted from the nonlinear library functions. Therefore, the accuracy of the SINDy framework strongly depends upon the selection of nonlinear candidate functions and their ability to capture the underlying flow physics. Improper construction of the nonlinear library may significantly affect the prediction accuracy.

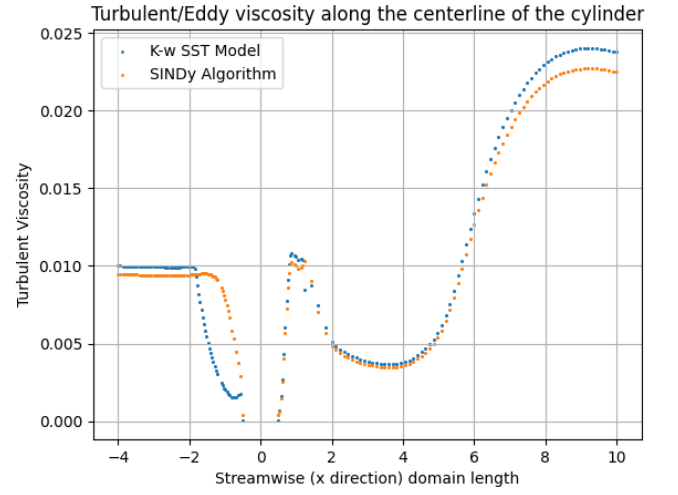
Although the model performed satisfactorily in most regions of the computational domain, discrepancies were observed in the wake region where the flow is highly separated and dominated by low pressure structures. In these regions, SINDy showed comparatively lower prediction accuracy for turbulent viscosity.

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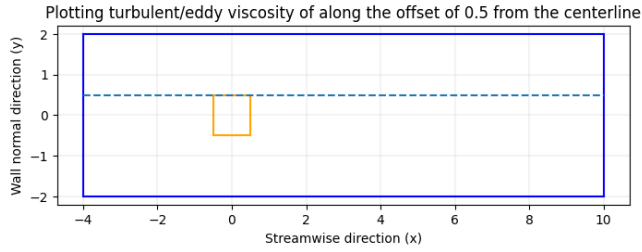


(a)

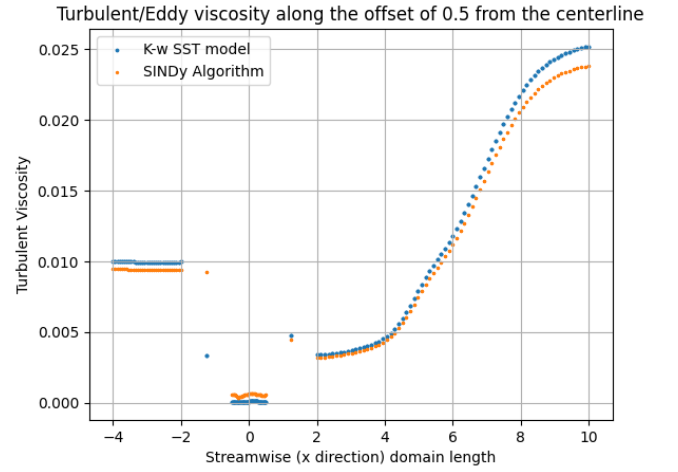


(b) Scatter plot for SST and SINDy

FIGURE 5: Comparison of turbulent viscosity contours obtained using SST and SINDy models along the centerline from the cylinder.

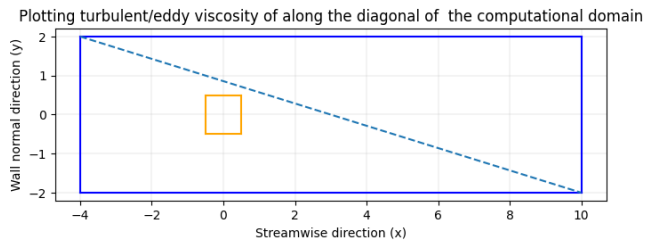


(a)

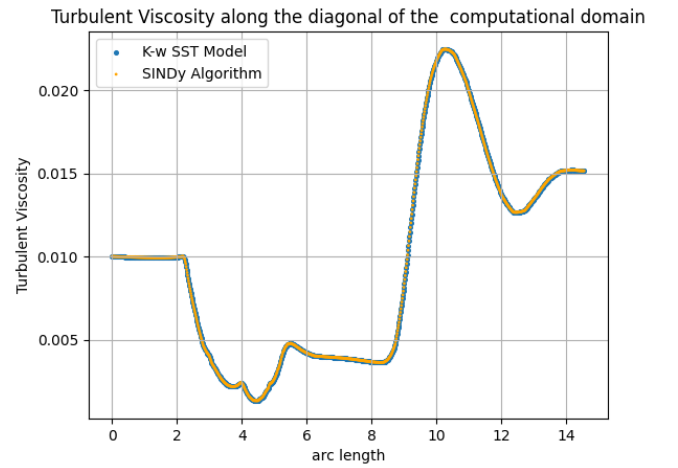


(b) Scatter Plot for SST and SINDy

FIGURE 6: Comparison of turbulent viscosity contours obtained using SST and SINDy models along the offset of 0.5 from the centerline

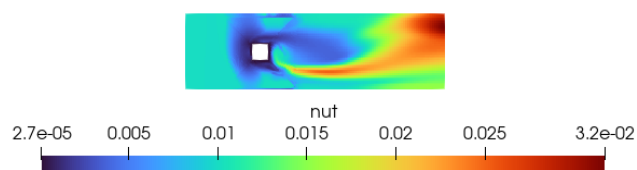


(a)

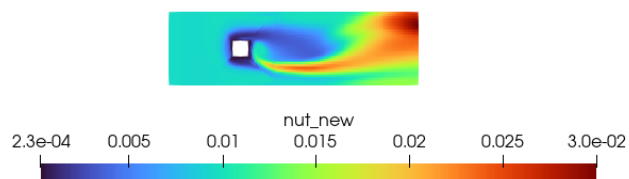


(b) Scatter plot of SSst and SINDy

FIGURE 7: Comparison of turbulent viscosity contours obtained using SST and SINDy models along diagonal of computational domain.



(a) SST turbulence viscosity contour



(b) SINDy turbulence viscosity contour

FIGURE 8: Comparison of turbulent viscosity contours obtained using SST and SINDy models.